

Highlights

- Four top-taggers (BDT, CNN, ParticleNet, Particle Transformer) compared under one pipeline.
- Data-efficiency sweep reveals a BDT-to-transformer crossover as training data grows.
- Ranking and calibration are decoupled: the CNN is worst-calibrated despite mid ranking.
- Temperature scaling fixes all models except the structurally miscalibrated CNN.
- Fully reproducible on a single free cloud GPU; code and models released openly.

Data efficiency and probability calibration of deep jet-tagging architectures on the 14 TeV top-quark benchmark

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Abstract

Top tagging — identifying the hadronic decays of highly boosted top quarks — is a standard benchmark for machine learning in collider physics. We compare four classifiers on the public Top-Quark Tagging Reference Dataset of 14 TeV proton–proton collisions: a gradient-boosted decision tree (BDT) on engineered jet-substructure variables, a convolutional neural network (CNN) on jet images, the ParticleNet graph network, and the Particle Transformer. Rather than chasing a new record area under the curve (AUC), we study two properties of direct relevance to downstream analyses: *data efficiency* and *probability calibration*. Using a fully reproducible pipeline trained on a single free cloud GPU, we sweep the training-set size from 5×10^3 to 5×10^4 jets and measure test AUC, background rejection at fixed signal efficiency, the Expected Calibration Error (ECE), and the effect of temperature scaling. The Particle Transformer attains the best discrimination ($\text{AUC} = 0.978$), reproducing the published ordering, but the data-efficiency sweep reveals a clear crossover: the BDT is strongest at the smallest training size and is only overtaken by the deep models as data grows. The calibration study shows that ranking power and calibration are decoupled — the jet-image CNN is the worst-calibrated model ($\text{ECE} = 0.078$) despite a mid-pack ranking, whereas the top-ranked transformer is comparatively well-calibrated, and a single-parameter temperature rescaling corrects every model except the CNN. The complete pipeline, trained models and documentation are released as open

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source.

Keywords: top tagging, jet substructure, deep learning, ParticleNet, Particle Transformer, data efficiency, probability calibration

1. Introduction

1.1. Physics context

The top quark is the heaviest known elementary particle, with a mass $m_t \simeq 172.76$ GeV [1]. Because this mass exceeds the hadronisation scale, the top decays as a bare quark, almost always through $t \rightarrow Wb$; in the fully hadronic channel the W decays as $W \rightarrow q\bar{q}'$, giving $t \rightarrow Wb \rightarrow q\bar{q}'b$. When the parent top carries a transverse momentum $p_T \gg m_t$ it is *boosted*: its three decay quarks are produced collinearly and, after parton showering and hadronisation, merge into one large-radius jet reconstructed by sequential-recombination algorithms such as anti- k_t [2, 3, 4]. Such a “top jet” differs from an ordinary jet initiated by a light quark or gluon (a QCD jet) in three measurable ways: it is more massive (mass $\sim m_t$), it shows a three-pronged rather than one-pronged radiation pattern, and it carries a higher constituent multiplicity. Exploiting these differences to separate top jets from the overwhelmingly more abundant QCD background — *top tagging* — is one of the canonical classification problems of LHC physics and a central tool in searches for physics beyond the Standard Model [5, 6].

1.2. From engineered features to deep learning

Classical taggers compress the jet into a handful of physically motivated *substructure observables*: the jet mass, the N -subjettiness ratios τ_{21}, τ_{32} [7], and energy-correlation functions such as C_2 and D_2 [8, 9]. Fed to a boosted decision tree (BDT), these provide a strong, interpretable baseline. The deep-learning era replaced hand-crafted features with representations learned directly from the constituents. Three paradigms followed in succession: pixelating the jet into a calorimeter-like image for a convolutional neural network (CNN) [10, 11]; treating the jet as an unordered “particle cloud” processed by a permutation-invariant graph network such as ParticleNet [12], built on the Dynamic Graph CNN [13]; and, most recently, modelling pairwise constituent interactions with self-attention in the Particle Transformer [14, 15]. Related set, graph and Lorentz-equivariant architectures continue to advance the field

[16, 17, 18, 19]. On the standard reference dataset [20], a community comparison [21] established the now-canonical ordering, with the leading deep models exceeding an area under the receiver operating characteristic curve (AUC) of 0.98.

1.3. Motivation and contributions

The published literature is almost entirely concerned with the *best achievable* discrimination, quoted at the full training-set size of 1.2×10^6 jets and on large GPU clusters. Two properties of direct practical importance are rarely studied together. The first is *data efficiency*: generating labelled Monte-Carlo training samples is computationally expensive, so how steeply each architecture’s performance depends on the amount of training data — and whether the ranking of the models changes in the small-data regime — is decisive for groups without large simulation budgets. The second is *probability calibration*: a high AUC certifies only that a model *ranks* signal above background, whereas many downstream analyses use the classifier score as a *probability* (for instance in a profile-likelihood fit), which requires the score to be calibrated. While calibration is routinely examined in the machine-learning literature [22, 23], it is seldom reported for jet taggers.

This paper addresses both questions on the public reference dataset using a single, reproducible pipeline that runs end-to-end on free hardware. Our contributions are:

1. a unified, open-source pipeline that trains and evaluates the BDT, CNN, ParticleNet and Particle Transformer under identical preprocessing, data splits and random seeds;
2. a data-efficiency study across a decade of training-set sizes (5×10^3 to 5×10^4 jets), revealing a clear crossover between the BDT and the deep models; and
3. a probability-calibration study, using reliability diagrams, the Expected Calibration Error and temperature scaling, showing that ranking power and calibration are decoupled, with practical recommendations for downstream use.

The remainder of the paper is organised as follows. Section 2 describes the dataset and preprocessing; Section 3 sets out the four models and the evaluation methodology with their governing equations; Section 4 presents the headline benchmark, the data-efficiency sweep and the calibration analysis; and Section 5 concludes.

2. Dataset and preprocessing

We use the public Top-Quark Tagging Reference Dataset [20, 21], 2×10^6 jets from 14 TeV pp collisions generated with PYTHIA 8 [24] and a DELPHES fast detector simulation [25], clustered with the anti- k_t algorithm [2, 3] at $R = 0.8$ and $550 < p_T < 650$ GeV. For tractability we draw stratified subsamples of up to 5×10^4 jets per split and use the leading 100 constituents per jet.

For each constituent we form the feature vector $(p_T, \eta, \phi, E, \Delta\eta, \Delta\phi, \log p_T)$ relative to the p_T -weighted jet axis, standardised on the training split. For the BDT we compute thirteen engineered observables (jet mass, p_T , η , multiplicity, width, leading/sub-leading p_T fractions, $\tau_1, \tau_2, \tau_3, \tau_{21}, \tau_{32}$ [7], and C_2 [8, 9]); their distributions (Fig. 1) behave exactly as physics expects — the top mass peaks at m_t and τ_{32} separates the three-prong signal. For the CNN we bin constituents into $40 \times 40 \times 3$ jet images [10]; the averaged images (Fig. 2) show the three-prong topology in the signal–background difference. For the Particle Transformer we add four pairwise features $(\ln \Delta R, \ln k_T, z, \Delta\eta)$ following Qu et al. [14].

3. Methodology

We treat top tagging as binary classification: each jet $\mathbf{x}$ is mapped by a model f_θ to a score $\hat{y} = f_\theta(\mathbf{x}) \in [0, 1]$ estimating the probability that the jet is a top jet. All four models are trained against the binary cross-entropy loss

$$\mathcal{L}(\theta) = -\frac{1}{N} \sum_{i=1}^N \left[y_i \log \hat{y}_i + (1 - y_i) \log(1 - \hat{y}_i) \right], \quad (1)$$

where $y_i \in \{0, 1\}$ is the truth label and the neural models produce $\hat{y}_i = \sigma(z_i)$ from a logit z_i through the logistic sigmoid $\sigma(z) = 1/(1 + e^{-z})$.

3.1. Boosted decision tree (BDT)

The baseline is a gradient-boosted decision-tree ensemble trained with XGBOOST [26] on the thirteen engineered observables. The prediction is an additive ensemble of K regression trees $f_k \in \mathcal{F}$,

$$z_i = \sum_{k=1}^K f_k(\mathbf{x}_i), \quad (2)$$

Jet Substructure Variables: Top vs QCD

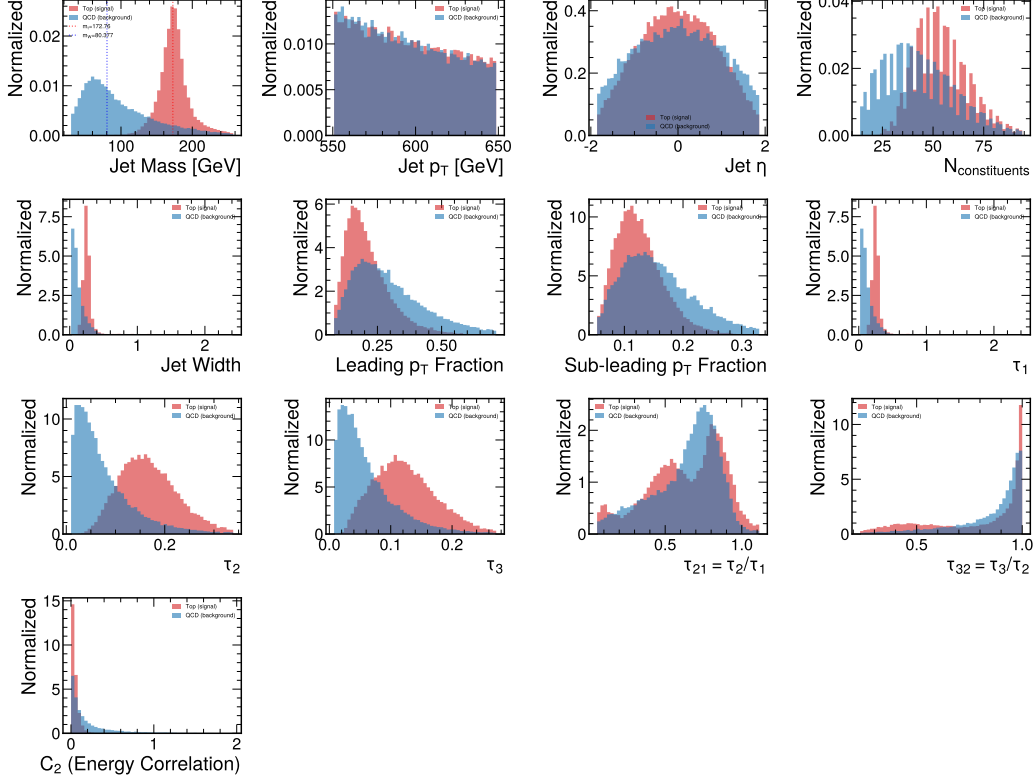


Figure 1: Engineered jet-substructure observables for top (red) and QCD (blue) jets on the test set.

built greedily by minimising the regularised objective

$$\mathcal{O} = \sum_i \ell(y_i, \hat{y}_i) + \sum_k \Omega(f_k), \quad \Omega(f) = \gamma T + \frac{1}{2} \lambda \|\mathbf{w}\|^2, \quad (3)$$

where ℓ is the logistic loss, T is the number of leaves, $\mathbf{w}$ the leaf weights, and γ, λ control complexity. At each boosting round a new tree is fit to the second-order Taylor expansion of $\mathcal{O}$, which yields fast, accurate splits. Among the inputs, the N -subjettiness ratios [7]

$$\tau_N = \frac{1}{d_0} \sum_k p_{T,k} \min(\Delta R_{1,k}, \dots, \Delta R_{N,k}), \quad \tau_{21} = \frac{\tau_2}{\tau_1}, \quad \tau_{32} = \frac{\tau_3}{\tau_2}, \quad (4)$$

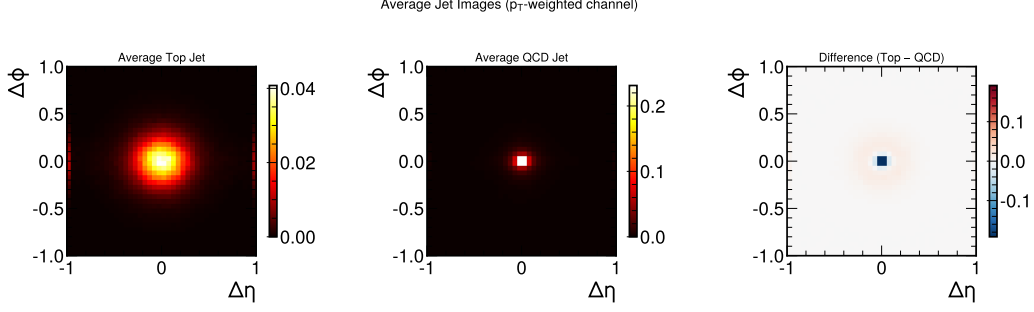


Figure 2: Averaged p_T -weighted jet images: top (left), QCD (centre) and their difference (right).

($d_0 = \sum_k p_{T,k} R_0$, $\Delta R_{a,k}$ the angular distance from constituent k to axis a) measure compatibility with an N -pronged hypothesis and are, with the jet mass, the most discriminating features.

3.2. Convolutional neural network (CNN)

The CNN acts on the $40 \times 40 \times 3$ jet image I , applying learnable kernels W in the $(\Delta\eta, \Delta\phi)$ plane,

$$(I * W)_{ij}^{c'} = \sum_c \sum_{a,b} W_{a,b}^{c',c} I_{i+a,j+b}^c, \quad (5)$$

interleaved with batch normalisation and ReLU nonlinearities $\text{ReLU}(u) = \max(0, u)$. We use a ResNet-style stack [27] in which each block learns a residual mapping

$$\mathbf{h}_{\ell+1} = \text{ReLU}(\mathbf{h}_\ell + \mathcal{F}(\mathbf{h}_\ell; W_\ell)), \quad (6)$$

easing optimisation of deeper networks. Global average pooling and a fully connected head produce the logit z .

3.3. ParticleNet (graph network)

ParticleNet [12] treats the jet as an unordered point cloud $\{\mathbf{x}_i\}$ and is built from EdgeConv blocks [13]. For each particle i , a k -nearest-neighbour graph is constructed in coordinate space and the feature is updated by a symmetric aggregation over its neighbourhood $\mathcal{N}(i)$,

$$\mathbf{x}'_i = \max_{j \in \mathcal{N}(i)} h_\Theta(\mathbf{x}_i, \mathbf{x}_j - \mathbf{x}_i), \quad (7)$$

where h_Θ is a shared multilayer perceptron and the element-wise max makes the operation permutation-invariant. The graph is recomputed in the learned feature space at each block (a “dynamic” graph), so successive layers capture progressively larger-scale structure; a final global pooling and dense head give z .

3.4. Particle Transformer

The Particle Transformer [14] applies multi-head self-attention [15] to the constituent sequence. With queries, keys and values Q, K, V obtained by linear projection, a standard attention head computes

$$\text{Attention}(Q, K, V) = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}}\right) V, \quad (8)$$

with d_k the key dimension. The key physics ingredient is a learned *interaction bias*: the four pairwise features $\mathbf{u}_{ij} = (\ln \Delta R_{ij}, \ln k_{T,ij}, z_{ij}, \Delta \eta_{ij})$ are mapped by an MLP to a per-head additive term $U_{ij} = \phi(\mathbf{u}_{ij})$ that is injected into the attention logits,

$$A = \text{softmax}\left(\frac{QK^\top}{\sqrt{d_k}} + U\right) V, \quad (9)$$

so that physically close, hard splittings are emphasised. A learnable class token aggregates the sequence and its representation is mapped to the logit z ; its attention weights provide an interpretability handle (Section 4).

3.5. Training

The neural models minimise Eq. (1) with Adam/AdamW [28, 29] (learning rates 10^{-3} – 10^{-4}), gradient clipping, a plateau learning-rate scheduler and early stopping on the validation AUC, implemented in PYTORCH [30] with NUMPY [31], SCIKIT-LEARN [32] and MATPLOTLIB [33]. Training used a free Google Colaboratory T4 GPU; the pipeline checkpoints every epoch and resumes automatically.

3.6. Evaluation metrics

We report the AUC, the accuracy at a score threshold of 0.5, and the background rejection $1/\varepsilon_B$ at fixed signal efficiencies $\varepsilon_S = 0.5, 0.3$, the figure of merit most relevant to physics analyses; AUC uncertainties are obtained by bootstrap resampling of the test set. Calibration is quantified by the Expected Calibration Error [22], which partitions the predictions into M

Table 1: Headline test-set performance (5×10^4 -jet training run).

Model	AUC	Acc.	$1/\varepsilon_B^{0.5}$	$1/\varepsilon_B^{0.3}$
BDT (XGBoost)	0.970	0.911	93	287
CNN (jet images)	0.972	0.915	111	413
ParticleNet	0.976	0.920	130	413
Particle Transformer	0.978	0.925	136	476

equal-width score bins B_m and averages the gap between confidence and accuracy,

$$\text{ECE} = \sum_{m=1}^M \frac{|B_m|}{N} |\text{acc}(B_m) - \text{conf}(B_m)|. \quad (10)$$

As a corrective we apply *temperature scaling*: a single scalar $T > 0$ is fit on the validation set by minimising the negative log-likelihood of the rescaled scores $\hat{y} = \sigma(z/T)$, and we report the ECE before and after rescaling. $T > 1$ indicates an over-confident model whose probabilities are softened by the rescaling.

4. Results

4.1. Headline benchmark

Table 1 and Fig. 3 report the headline performance. The ordering Particle Transformer (0.978) > ParticleNet (0.976) > CNN (0.972) > BDT (0.970) reproduces the community benchmark [21]; absolute values are slightly below the published saturation values because we train on a subset rather than the full 1.2×10^6 jets.

4.2. Data efficiency

Re-training each model at $N_{\text{train}} \in \{5, 10, 25, 50\} \times 10^3$ jets yields the learning curves of Fig. 4. The BDT is the most data-efficient (AUC 0.961 $\rightarrow$ 0.968 across the range), whereas the deep models gain strongly and *cross over*: the Particle Transformer is the weakest model at 5×10^3 jets (AUC 0.949) but the strongest at 5×10^4 (0.974). Under a tight Monte-Carlo budget the simple, interpretable BDT is therefore competitive or preferable; the attention model only realises its advantage with sufficient data, consistent with its near- 10^6 -jet saturation [14].

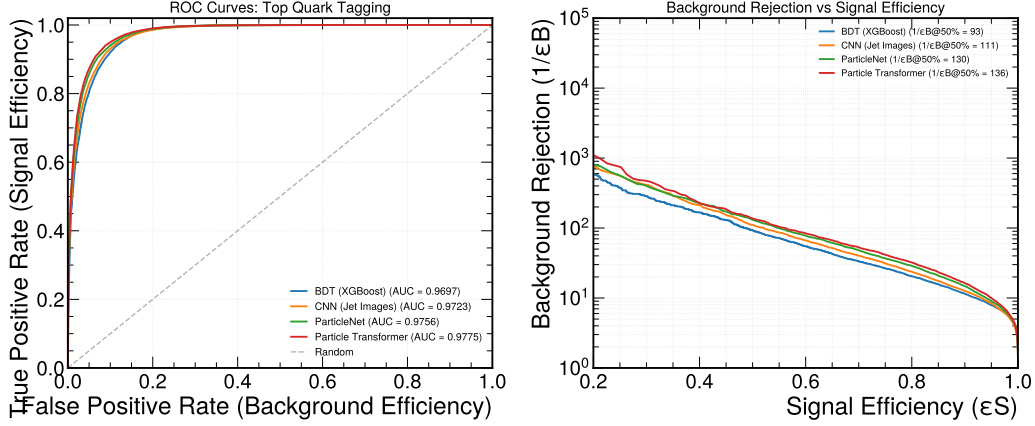


Figure 3: ROC curves (left) and background rejection vs. signal efficiency (right).

Table 2: Test AUC versus training-set size N_{train} . Uncertainties are one standard deviation from a 200-sample bootstrap of the test set. The Particle Transformer is weakest at 5×10^3 jets and strongest at 5×10^4 , crossing the BDT and ParticleNet.

Model	5×10^3	10^4	2.5×10^4	5×10^4
BDT (XGBoost)	0.961 ± 0.001	0.964 ± 0.001	0.967 ± 0.001	0.968 ± 0.001
CNN (jet images)	0.947 ± 0.001	0.953 ± 0.001	0.968 ± 0.001	0.968 ± 0.001
ParticleNet	0.950 ± 0.001	0.955 ± 0.001	0.966 ± 0.001	0.971 ± 0.001
Particle Transformer	0.949 ± 0.001	0.952 ± 0.001	0.966 ± 0.001	0.974 ± 0.001

4.3. Probability calibration

Following Guo et al. [22] we compute the Expected Calibration Error (ECE) over 15 score bins and fit a single temperature T on the validation set. Figure 5 and Table 3 show that all models are mildly over-confident ($T > 1$) but to very different degrees. Strikingly, ranking and calibration are *decoupled*: the jet-image CNN is the worst-calibrated by an order of magnitude ($\text{ECE} = 0.078$) despite ranking third on AUC, whereas the top-ranked Particle Transformer is comparatively well-calibrated ($\text{ECE} = 0.012$). Temperature scaling brings the BDT, ParticleNet and transformer to $\text{ECE} \lesssim 0.008$ but only halves the CNN’s error (to 0.064): the CNN’s miscalibration is structural and would require a richer recalibration such as isotonic regression [23].

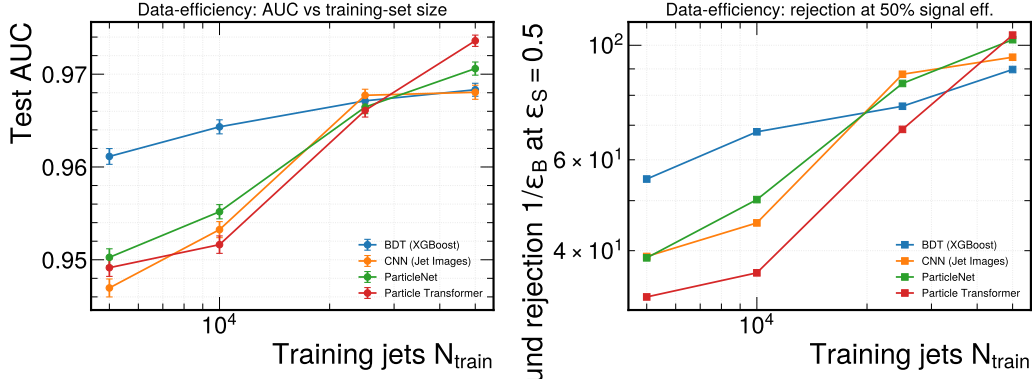


Figure 4: Data-efficiency curves: test AUC (left) and background rejection at $\epsilon_S = 0.5$ (right) vs. training-set size.

Table 3: Calibration at the 5×10^4 -jet endpoint: ECE, fitted temperature T , and ECE after rescaling.

Model	ECE	T	ECE_T
BDT (XGBoost)	0.007	1.04	0.006
CNN (jet images)	0.078	1.54	0.064
ParticleNet	0.008	1.05	0.008
Particle Transformer	0.012	1.12	0.007

4.4. Interpretability

The class-token attention of the Particle Transformer (Fig. 6) concentrates at small angular distance ΔR for QCD jets (a single core) but spreads to large ΔR for top jets, reflecting the broader multi-prong energy flow of the top decay. The BDT feature importances are dominated by the jet mass, τ_{32} and multiplicity, the physically expected discriminators.

5. Conclusions

We have presented a reproducible comparison of four top-tagging architectures focused on data efficiency and probability calibration. The literature ordering is reproduced, but two practical messages emerge. First, data efficiency separates the models: the BDT is best in the small-data regime, with the deep models overtaking it only as data grows. Second, ranking and calibration are decoupled — the best-ranked transformer is well-calibrated, while

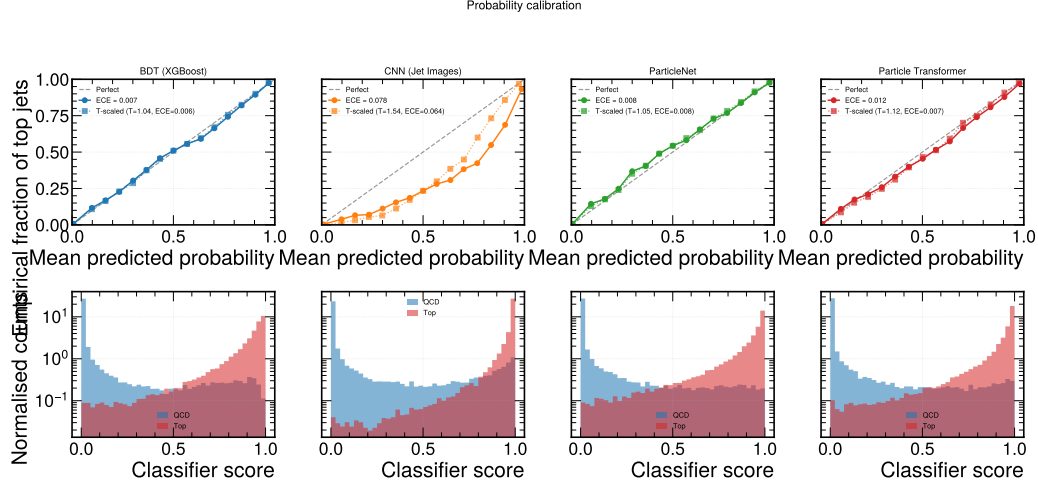


Figure 5: Reliability diagrams (top) and score histograms (bottom). Solid: raw scores; dotted: temperature-scaled. The CNN deviates most.

the CNN, mid-pack on AUC, is the worst-calibrated and is not fully corrected by temperature scaling. We recommend reporting temperature-scaled probabilities by default and treating CNN scores with particular caution. Natural extensions are to scale the sweep to the full dataset, to study robustness under pile-up, and to compare non-parametric recalibration.

Code and data availability

The complete pipeline, trained models, and documents are openly available at <https://github.com/ramc77/fyp-jet-tagging>. The dataset is the public Top-Quark Tagging Reference Dataset [20] (<https://zenodo.org/records/2603256>).

CRedit authorship contribution statement

Bibi Naifa: Methodology, Software, Investigation, Writing – original draft. **Mah Lacka:** Software, Validation, Visualization, Writing – original draft. **Kanwal Gul:** Formal analysis, Data curation, Writing – review & editing. **Firdos:** Investigation, Validation, Writing – review & editing. **Ram Chand:** Conceptualization, Supervision, Project administration, Methodology, Writing – review & editing.

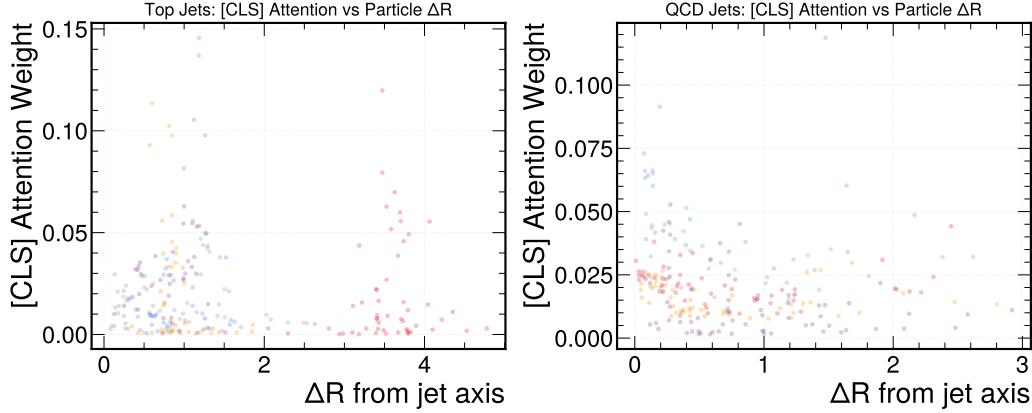


Figure 6: Class-token attention vs. constituent ΔR for top jets (left) and QCD jets (right).

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgements

We thank the maintainers of the Top-Quark Tagging Reference Dataset and the open-source scientific-Python and Google Colaboratory ecosystems. This research received no specific grant from any funding agency.

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